## Submission and policies

- **Report:** upload one PDF to Gradescope under "Assignment 2", including your answers and all images generated by your code. Support interpretations with specific values or features in your results; concise annotations are enough. See the Gradescope submission instructions for help.
- **Code:** upload all source files and `results.json` to "Assignment 2 Code", individually or in one ZIP archive. Use Python, C/C++, Julia, or another reasonable language; comment your code so we can follow and run it.
- **Grading:** 40 required points plus 10 bonus points for E. The autograder awards 16 required points and 4 bonus points; the report earns the rest.
- **Collaboration:** follow the syllabus policy on all problems.
- **AI use appendix:** end your report with the tools used and their purpose, your prompts (or a conversation link), and what the tools got wrong and how you caught it, as required by the syllabus. "I did not use AI tools on this assignment" is a complete answer.
- **Oral Checkpoint 2:** Lab 12, Thursday, November 5. Bring your submitted code on a laptop, with AI assistants closed. We may select a result or code segment, ask you to predict a small change chosen during the meeting, and have you make and check it. Be ready to connect the result to the relevant code; bonus E is not required for the checkpoint.

## Numerical conventions

- **Grid:** $N_D + 1 = 601$ points from $x_0 = -4$ to $x_{N_D} = 4$, with $N_D = 600$ and $\Delta x = 8/600$. Only B(ii)'s convergence checks change this grid; all JSON results use the original grid.
- **Time step:** $\epsilon = \Delta t = \pi/128$.
- **Integrals:** use $\int f(x)\,dx \approx \Delta x \sum_{i=0}^{N_D} f(x_i)$, including both endpoints, for propagation, normalization, expectation values, and overlaps.
- **Tunneling time:** $t_{\text{tunnel}}$ is the time of the first major minimum of $\langle x \rangle(t)$ in the slow right-to-left oscillation: its first maximum leftward displacement. Do not use the first zero crossing or a small minimum from faster oscillations.

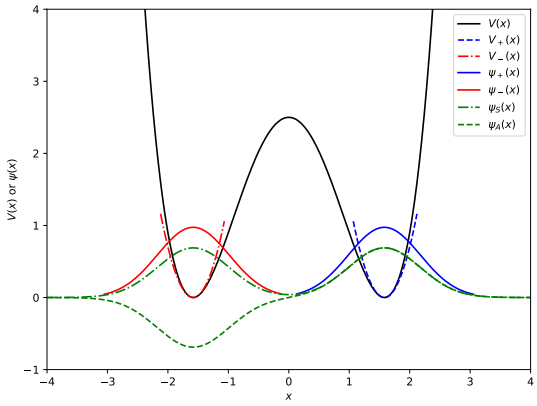
# 1 Tunneling in the Double Potential Well [40 Points + 10 Bonus Points]

*Relevant materials: Lectures 10–11*

**Potential.** For a particle at position $x$, set $m = \hbar = 1$ and use

$$V(x) = \alpha x^4 - 2x^2 + \frac{1}{\alpha}, \tag{1}$$

The minima are at $x = \pm x_{\min}$, where $x_{\min} = 1/\sqrt{\alpha}$ and $V(\pm x_{\min}) = 0$. The barrier height is $V(0) = 1/\alpha$; the figure uses $\alpha = 0.4$.




**Your parameter.** Let $d$ be the last three digits (000–999) of your Triton Student Number (TSN: nine digits starting with 2). Use

$$\alpha = 0.3 + \frac{d}{5000}, \tag{2}$$

so $0.3 \le \alpha < 0.5$. Examples: TSN 212345678 gives $\alpha = 0.3 + 678/5000 = 0.4356$; TSN 212345123 gives $\alpha = 0.3246$. State $\alpha$ at the top of your report and use it unless other values are specified.

**Harmonic approximation.** Since $V''(\pm x_{\min}) = 8$ for every $\alpha$, the approximate wells near each minimum are

$$V_+(x) = 4(x - x_{\min})^2, \tag{3}$$
$$V_-(x) = 4(x + x_{\min})^2, \tag{4}$$

with $\omega = 2\sqrt{2}$ and ground states

$$\psi_\pm(x) = \left(\frac{\omega}{\pi}\right)^{1/4} \exp\left(-\frac{\omega}{2}(x \mp x_{\min})^2\right), \tag{5}$$

Their symmetric and antisymmetric combinations,

$$\psi_S(x) = \frac{1}{\sqrt{2}}\left(\psi_+(x) + \psi_-(x)\right), \qquad \psi_A(x) = \frac{1}{\sqrt{2}}\left(\psi_+(x) - \psi_-(x)\right) \tag{6}$$

approximate the ground state $\phi_0$ and first excited state $\phi_1$, whose energy gap is $\Delta E = E_1 - E_0$.

**Problem A [10 points]: Path-integral tunneling**

Start in the right well, $\psi(x,0) = \psi_+(x)$, and propagate one time step at a time with

$$\tilde{\mathcal{K}}(x_b, \epsilon; x_a, 0) = \exp\left[i\left(\frac{1}{2}\frac{(x_b - x_a)^2}{\epsilon} - V\left(\frac{x_a + x_b}{2}\right)\epsilon\right)\right], \tag{7}$$

where $x_a$ and $x_b$ are the initial and final positions. The autograder requires this midpoint potential. The propagator omits its normalization factor, so normalize after every step:

$$\psi(x,t) \leftarrow \frac{\psi(x,t)}{\sqrt{\int |\psi(x,t)|^2\, dx}}. \tag{8}$$

(i) **Run and plot.** Follow one full slow oscillation, from right to left and back. Plot $\langle x\rangle(t)$ and mark the measured $t_{\text{tunnel}}$. You may estimate the run duration with $2\pi/\Delta E$ from B, but do not substitute $\pi/\Delta E$ for the measured time.

(ii) **Read your result.** Give the time and value of your chosen major minimum. Mark a faster fluctuation if one is resolved, and distinguish it from the slow transfer. Compare with D's two-state curve: which feature of your results supports your account of any departures from a single sinusoid?

**Problem B [10 points]: Numerical eigenstates**

Find the two lowest eigenvalues $E_0, E_1$ and eigenfunctions $\phi_0, \phi_1$ of $\hat{H} = -\frac{1}{2}\frac{\partial^2}{\partial x^2} + V(x)$ (Lecture 11). Use the $(N_D+1)\times(N_D+1)$ centered three-point matrix

$$H_{ii} = \frac{1}{\Delta x^2} + V(x_i), \qquad H_{i,i+1} = H_{i+1,i} = -\frac{1}{2\Delta x^2}, \tag{9}$$

with all other entries zero.

**Endpoints:** retain all $N_D + 1$ unknowns, including both endpoints. Omit the outside neighbor in the first and last rows: this sets $\psi = 0$ at $x_0 - \Delta x$ and $x_{N_D} + \Delta x$, not at the endpoints themselves.

**Solver:** use shift-invert via `scipy.sparse.linalg.eigsh` with `sigma`, or your own shifted inverse power iteration. Since $V \geq 0$ gives $E_n > 0$, a shift $\sigma = 0$ finds the lowest states.

(i) **Validate first.** Compute $E_0, E_1$ at $\alpha = 0.4, 0.6$ and compare with Table I of K. Banerjee and S. P. Bhatnagar, "Two-well oscillator", Phys. Rev. D 18, 4767 (1978), shown in Lecture 11. Their $\lambda = \alpha/4$, so use $\lambda = 0.10, 0.15$. Report how many digits agree. Use the comparison and your convergence checks to discuss what may limit the agreement.

(ii) **Check convergence at your $\alpha$.** Report $E_0, E_1, \Delta E$ on the original grid, then test:

| Grid | Interval | $N_D$ | $\Delta x$ |
|---|---|---|---|
| Original | $[-4, 4]$ | 600 | 8/600 |
| Half spacing | $[-4, 4]$ | 1200 | 8/1200 |
| Larger box | $[-6, 6]$ | 900 | 8/600 |

Tabulate each gap and its change from the original result. Use those changes to assess sensitivity to grid spacing and box size, and justify the digits you quote. Identify the code that sets the grid spacing and updates the Hamiltonian. Use the original-grid results thereafter, including in `results.json`.

(iii) **Plot eigenfunctions.** Normalize with $\Delta x \sum_i |\phi_n(x_i)|^2 = 1$. Choose $\phi_0$ positive and the right-hand lobe of $\phi_1$ positive. Plot both together, with $V(x)$ in a separate panel sharing the position axis.

**Problem C [10 points]: Harmonic-approximation energies**

Estimate the energies using $\psi_S, \psi_A$:

$$E_S = \frac{\int \psi_S^*(x)\, \hat{H}\psi_S(x)\, dx}{\int |\psi_S(x)|^2\, dx}, \qquad E_A = \frac{\int \psi_A^*(x)\, \hat{H}\psi_A(x)\, dx}{\int |\psi_A(x)|^2\, dx}. \tag{10}$$

Use B's Hamiltonian matrix and grid sums. Keep the denominators: the slight overlap of $\psi_+$ and $\psi_-$ leaves $\psi_S, \psi_A$ unnormalized, which matters at this precision.

(i) **Compare energies.** Compare $E_S, E_A, E_A - E_S$ with B's $E_0, E_1, \Delta E$. Report their relative errors (e.g., $|E_S - E_0|/|E_0|$). Use the signed errors $E_S - E_0$ and $E_A - E_1$ to account for the gap error. Which is more accurate in relative terms: the energies or their gap?

(ii) **Compare wave functions.** Normalize $\psi_S$, then overlay $|\psi_S|$ and $|\phi_0|$ with a logarithmic vertical scale. Mark a discrepancy in the barrier region and relate it to the limits of the local harmonic approximation. Use your energy errors from (i) to show why the relative error in the gap can be large despite smaller relative errors in the energies.

**Problem D [10 points]: Compare two evolution methods**

Use the same right-well initial state for both methods: $\psi(x,0) = (\phi_0(x) + \phi_1(x))/\sqrt{2}$, with B's signs. Set $T = 2\pi/\Delta E$ (one full oscillation) and evolve to the first time step at or after $T$.

(i) **From the eigenstates:**

$$\psi(x,t) = \frac{1}{\sqrt{2}} \left( e^{-iE_0 t/\hbar}\, \phi_0(x) + e^{-iE_1 t/\hbar}\, \phi_1(x) \right). \tag{11}$$

Show that $\langle x \rangle(t) = x_{01} \cos(\Delta E\, t/\hbar)$ with $x_{01} = \int \phi_0(x)\, x\, \phi_1(x)\, dx$, and hence that $t_{\text{tunnel}} = \pi\hbar/\Delta E$ for this initial state. Identify where normalization and symmetry enter your derivation.

(ii) **With the path integral:** reuse A's code with the two-state initial wave function above.

(iii) **Compare.** Overlay both $\langle x \rangle(t)$ curves. Evaluate them at every time step and report

$$\max_{0 \le n\epsilon \le T} \left| \langle x \rangle_{\text{PI}}(n\epsilon) - \langle x \rangle_{\text{eig}}(n\epsilon) \right|. \tag{12}$$

Overlay both probability densities at the nearest steps to $0, t_{\text{tunnel}}, T$; label the methods and actual times. At the maximum discrepancy, report the time and both means. Use your curves or snapshots to identify a discrepancy pattern and a possible numerical cause. Propose one targeted check and predict its outcome if that cause is responsible; no extra run is required.

**Optional animation:** save both densities as a `.gif` or `.mp4`. This earns no extra points and is not required for full credit.

**Problem E [10 bonus points]: Test the two-state prediction** *(optional)*

D predicts $t_{\text{tunnel}} = \pi/\Delta E$ for $(\phi_0 + \phi_1)/\sqrt{2}$. Test this prediction for $\psi_+$, a localized state that also contains higher states. Use $\alpha = 0.30, 0.35, 0.40, 0.45, 0.50$.

(i) **Measure.** For each $\alpha$, compute $\Delta E$ with B's code and measure $t_{\text{tunnel}}$ with A's path integral, starting from $\psi_+$ and using the major-minimum definition.

(ii) **Compare.** Plot $t_{\text{tunnel}}$ against $\Delta E$ on logarithmic axes alongside $\pi/\Delta E$. How well does the prediction hold?

(iii) **Interpret.** Normalize $\psi_+$ on the grid and compute $|\langle\phi_0|\psi_+\rangle|^2 + |\langle\phi_1|\psi_+\rangle|^2$ for each $\alpha$: the fraction in the two lowest states. Compare two of your $\alpha$ values using their fractions, deviations from $\pi/\Delta E$, and faster oscillations in $\langle x\rangle(t)$. Do these results support a higher-state explanation? Distinguish that evidence from possible numerical errors in D and uncertainty in locating the major minimum.

## Autograded results

Have your code write `results.json` and upload it with the source files. Any language can write JSON; a Python example follows the field table.

- **Parameters:** use your $\alpha$ and the original $N_D = 600$ grid. Only E's two lists use other $\alpha$ values; omit those lists if you skip E.
- **Export times:** $t = n\epsilon$ for $n = 0, 64, 128, \ldots, 2048$ (33 values, ending at $16\pi$). These times do not define a full oscillation. Run A through at least $16\pi$ and longer if needed for a full slow oscillation. Evaluate D's eigenstate formula at all 33 times, even beyond its plotted oscillation.
- **Initial states:** `x_mean_pi` is A's Gaussian result. D's two-state path-integral curve goes in the report, not in that field.

| Field | Contents |
|---|---|
| `tsn` | Your TSN, e.g., `"212345678"` |
| `alpha` | Your $\alpha$ |
| `x_mean_pi` | Problem A: $\langle x\rangle$ at $t = n\epsilon$ for $n = 0, 64, 128, \ldots, 2048$ (33 values), path integral starting from $\psi_+$ |
| `t_tunnel` | Problem A: your estimate of $t_{\text{tunnel}}$ |
| `E0, E1` | Problem B: $E_0$ and $E_1$ |
| `phi0, phi1` | Problem B: $\phi_0(x_i)$ and $\phi_1(x_i)$ on the grid (601 values each), normalized and with the signs of Problem B |
| `E_S, E_A` | Problem C: $E_S$ and $E_A$ |
| `x_mean_eig` | Problem D(i): $\langle x\rangle$ from the eigenstates at the same 33 times as in Problem A |
| `delta_E_list` | Problem E (bonus): $\Delta E$ for $\alpha = 0.30, 0.35, 0.40, 0.45, 0.50$, in that order |
| `t_tunnel_list` | Problem E (bonus): your measured $t_{\text{tunnel}}$ for the same five values of $\alpha$ |

```
import json

results = {
    "tsn": "212345678",
    "alpha": alpha,
    "x_mean_pi": x_mean_pi.tolist(),
    "t_tunnel": float(t_tunnel),
    "E0": float(E0), "E1": float(E1),
    "phi0": phi0.tolist(), "phi1": phi1.tolist(),
    "E_S": float(E_S), "E_A": float(E_A),
    "x_mean_eig": x_mean_eig.tolist(),
    # Problem E (bonus); leave these two out if you skip it
    "delta_E_list": [float(dE) for dE in delta_E_list],
    "t_tunnel_list": [float(t) for t in t_tunnel_list],
}
with open("results.json", "w") as f:
    json.dump(results, f)
```

## Grading and feedback

- **Autograded points:** A: 4, B: 6, C: 3, D: 3; E: 4 bonus. The report earns the remaining points. The grader uses your assigned $\alpha$ except for E's fixed values.
- **Tolerances:** tunneling times have loose checks because reading the minimum is approximate. Other quantities have close checks: follow the stated grid, time step, propagator, normalization, and signs.
- **Feedback:** Gradescope immediately checks that it can read `results.json` and that $\alpha$ matches your TSN. Scores appear after the deadline; resubmit as often as you like before then.